CHAPTER 1

Workshop Overview & Elephant Mortality Analysis

Chhattisgarh 2021–2026: Trends, Causes & Recommendations

Chhattisgarh Forest Department | Technical Support: WII Dehradun

1. Workshop Background & Objectives

The **Workshop on Essentials for Mortality Investigation of Asian Elephant** was organised by the Chhattisgarh Forest Department with technical support from the Wildlife Institute of India (WII), Dehradun, and laboratory partners ICAR-IVRI, Bareilly and NDVSU, Jabalpur. The two-day programme (5–6 June 2026, Raigarh) brought together field officers and wildlife veterinarians to build capacity in systematic elephant necropsy, sample collection, disease diagnosis, and evidence-based mortality investigations.

84 Participants

2 Days Training

5 Expert Resource Persons

3 Partner Institutions

Workshop Objectives

Theme	Topics Covered
Biology & Ecology	Biology, behaviour, ecology of Asian elephants; population dynamics in CG
Pathology & Disease	Necropsy procedures, biosafety, sample collection, infectious & non-infectious diseases
Surveillance & Response	Elephant surveillance systems, disposal protocols, site remediation, evidence documentation

Resource Persons

Name	Institution	Expertise
Dr. A. B. Shrivastav	NDVSU, Jabalpur	Wildlife Pathology, Elephant PM Examination
Dr. Parag Nigam	WII, Dehradun	Elephant Ecology & Population Biology
Dr. Karikalan Mathesh	ICAR-IVRI, Bareilly	Veterinary Pathology, Sample Diagnostics
Dr. Tapendra Saini	WII, Dehradun	Wildlife Surveillance & Disease Monitoring
Dr. Chandra Prakash Sharma	WII, Dehradun	Conservation Biology & Field Investigation

2. Elephant Population Status in Chhattisgarh

Chhattisgarh has witnessed a remarkable expansion of its elephant population over the past two decades. Elephants, originally transient from Odisha and Jharkhand, have established resident breeding populations across two Forest Divisions: **Dharamjaigarh (DH)** and **Raigarh (RG)**. The population has grown from just 24 individuals in 2001 to an estimated **451 in 2026**, representing a Compound Annual Growth Rate (CAGR) of approximately 13%.

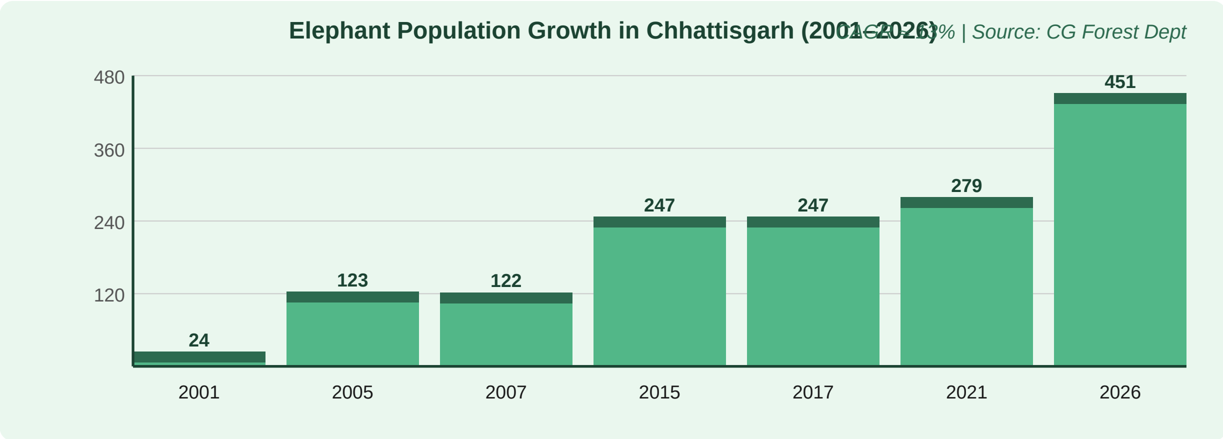


Figure 1: Elephant population growth in Chhattisgarh (2001–2026). Bilaspur Circle holds the highest concentration.

Year	Population Count	Remarks
2001	24	First recorded transient elephants from Odisha
2005	123	Rapid establishment of resident herds
2007	122	Stable — consolidation phase
2015	247	Continued range expansion into new areas
2017	247	Stable count; human-elephant conflict rising
2021	279	Post-COVID survey — increased mortality recorded
2026	~451	**Estimated — highest ever; detailed census ongoing**

3. Elephant Casualty Analysis (2021–2026)

TOTAL: 49 Elephant Deaths | DH Division: 27 | RG Division: 22 | 2025-26: 12 deaths (RECORD HIGH) — Immediate interve

Year-wise Elephant Casualties in Chhattisgarh (2021–2027*)

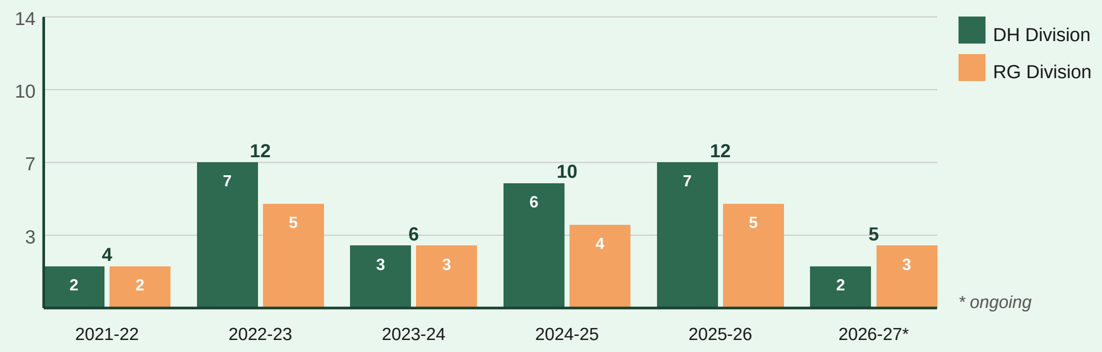


Figure 2: Year-wise elephant casualties by division (DH = Dharamjaigarh, RG = Raigarh). * 2026-27 data as of June 2026 (ongoing year).

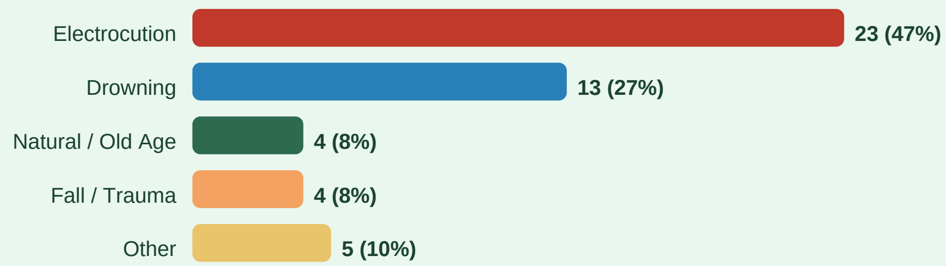
Year	DH Division	RG Division	Total	Remarks
2021-22	2	2	4	Baseline year — low mortality
2022-23	7	5	12	Surge in deaths — monsoon related
2023-24	3	3	6	Relatively low year
2024-25	6	4	10	Rising trend resumes
2025-26	7	5	**12**	**Record high — electrocution predominant**
2026-27*	2	3	5*	Ongoing year (as of June 2026)
TOTAL	**27**	**22**	**49**	**Cumulative 2021-2026**

Human Casualties in Human-Elephant Conflict

Alongside elephant deaths, the ongoing human-elephant conflict has resulted in **45 human deaths** during 2021–2026 (DH: 36 deaths | RG: 9 deaths). This underscores the urgency of conflict mitigation alongside mortality investigation capacity.

4. Cause of Death Analysis

Cause of Death Analysis (2021–2026, n=49)



Electrocution + Drowning = 74% — PREVENTABLE deaths

Figure 3: Cause of death distribution (n=49). Electrocution and drowning account for 74% of all deaths — both are largely preventable with targeted interventions.

Cause of Death	Count	Percentage	Key Mechanism
Electrocution	23	47%	Stray power lines, illegal electric fencing around crops
Drowning	13	27%	Calves unable to exit water bodies; steep banks, dams
Natural / Old Age	4	8%	Multi-molar loss, senility — natural attrition
Fall / Trauma	4	8%	Steep terrain, mine shafts, road/rail conflict
Other / Undetermined	5	10%	Toxicology pending, disease, inter-elephant conflict

5. Demographic Analysis of Casualties

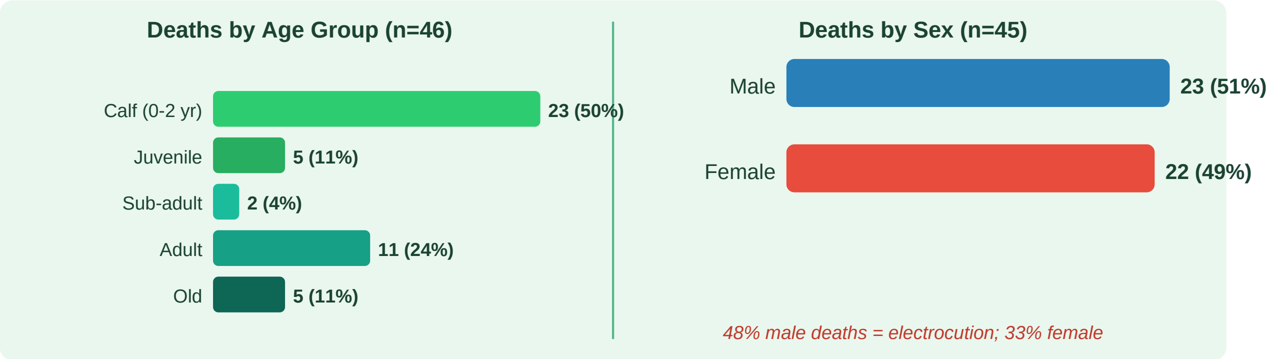


Figure 4: Age group (left) and sex (right) distribution of casualties (n=46 and n=45 respectively). Calves are disproportionately affected — 50% of all deaths.

Age Category	Count	%	Key Vulnerability
Calf (0–2 yr)	23	50%	Drowning (calves cannot exit steep banks), electrocution
Juvenile (2–5 yr)	5	11%	Separation from herd, foraging near crop fields
Sub-adult (5–15 yr)	2	4%	Dispersing males — increased exposure to power lines
Adult (15–40 yr)	11	24%	Electrocution in bulls; old cow mortality
Old (>40 yr)	5	11%	Natural senility, molar exhaustion, starvation

CRITICAL: Calves (0-2 yr) account for 50% of all deaths and 92% of all drowning deaths. Drowning prevention infrastructu

6. Seasonal Variation in Mortality

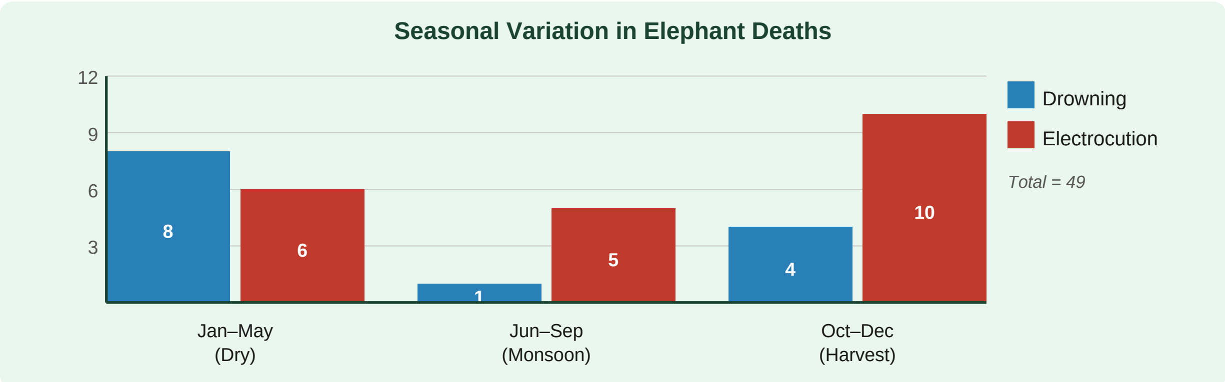


Figure 5: Seasonal distribution of deaths by cause. October is the single deadliest month (8 deaths). Oct–Dec harvest season accounts for 42% of annual deaths.

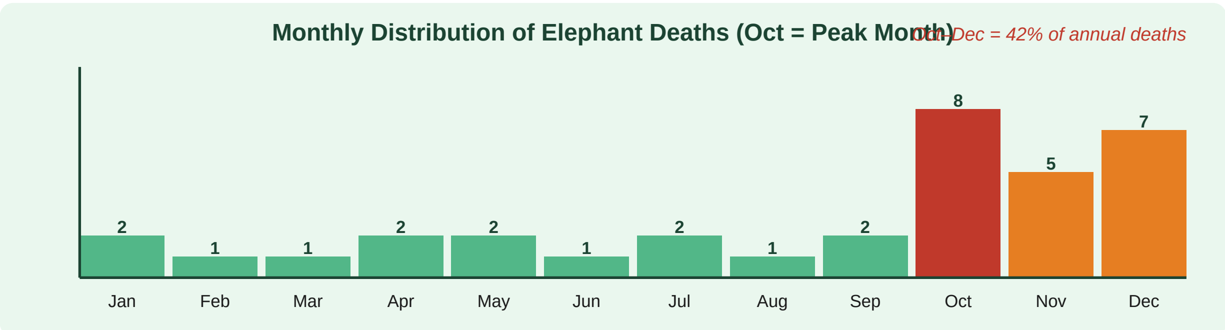


Figure 6: Monthly distribution of all elephant deaths (2021–2026 combined). October–December forms a clear mortality peak driven by illegal electric fencing during harvest.

Season	Months	Drowning	Electrocution	Key Driver
Dry Season	Jan–May	8	6	Low water levels — calves trapped in drying pools / tanks
Monsoon	Jun–Sep	1	5	Flooded rivers, reduced visibility of power lines
Harvest Season	**Oct–Dec**	**4**	**10**	**PEAK: Illegal electric fencing to protect paddy/maize crops**
TOTAL	—	**13**	**21***	*excludes 2 unconfirmed electrocution deaths

7. Range-wise Distribution of Deaths

Analysis of death locations reveals distinct spatial hotspots that require targeted management interventions. Two ranges account for 59% of all deaths and warrant immediate priority action.

Forest Range	Division	Elec.	Drown.	Other	Total	Status
Ghargoda	RG	8	5	2	**15**	**CRITICAL**
Chhal	DH	7	5	2	**14**	**CRITICAL**
Dharamjaigarh	DH	3	2	1	6	HIGH
Kharsia	RG	2	1	1	4	MODERATE
Tamnar	RG	2	1	1	4	MODERATE
Other Ranges	—	1	—	5	6	Monitor
TOTAL	—	**23**	**13**	**13**	**49**	—

Human Conflict Hotspots

Range	Human Deaths	Priority Status
Chhal	10	HIGHEST PRIORITY — targeted patrol + power audit
Borojh	9	HIGH — community early warning system needed
Lailungaan	8	HIGH — corridor mapping and fencing required
Total (45 deaths)	45	Comprehensive conflict mitigation plan required

8. Suspected Drowning — Post-Mortem Analysis

KEY FINDING: Drowning = 13 deaths (27% of total). 92% of drowning victims were calves. ARSENIC DETECTED in 2 drown

All 13 suspected drowning deaths underwent systematic post-mortem examination. Consistent PM findings confirmed asphyxia/drowning as the primary cause. Critically, diatom testing was positive in all tested cases. Two specimens from Tamnar Range (Dec 2025) returned positive for arsenic (File No. F.2-19/DI/NRC/2025-26/CWL), indicating possible water body contamination requiring environmental follow-up.

PM Finding	Observation	Significance
Frothy discharge	Present in ALL 13 cases	Hallmark of antemortem drowning — not seen in post-mortem drowning
Waterlogged lungs	Present in ALL 13 cases	Lung weight 3–5× normal; confirms inhalation of water
Water in trachea	Present in ALL 13 cases	Rules out post-mortem artefact — water aspirated while alive
GI water content	Large volume — ALL cases	Swallowed water while struggling — corroborates active drowning
Diatom test	**POSITIVE — all tested cases**	Diatoms in bone marrow = systemic dissemination = confirmed antemortem drowning
Arsenic	**DETECTED in 2 specimens**	Tamnar Range, Dec 2025 — environmental contamination investigation initiated

EEHV	NEGATIVE all cases	Viral haemorrhagic disease ruled out
Nitrate-Nitrite	NEGATIVE all cases	Fertiliser/industrial contamination ruled out
Heavy metals (Pb/Hg/Cd)	NEGATIVE all cases	Common industrial contaminants not implicated
Organochlorine/phosphate	NEGATIVE all cases	Pesticide poisoning excluded
HCN (Cyanide)	NEGATIVE all cases	Plant/industrial cyanide toxicity excluded

9. Case Study — Gurda Calf Drowning (01 June 2026)

Parameter	Details
Date	01 June 2026
Location	Gurda River sandbank, Kharsia Range, Raigarh Van Mandal
GPS Coordinates	22.0439°N, 83.1313°E
Victim	Male calf, estimated age 12–18 months
Discovery	Calf found separated from herd on exposed sandbank after overnight flooding
Circumstantial evidence	River flash flood event; steep eroded banks at GPS location; no other herd members present at recovery
PM Findings	Waterlogged lungs, frothy tracheal discharge, water in stomach (large volume), no external trauma, no injuries consistent with predation or electrocution
Diatom Test	POSITIVE — Naviculaceae spp. detected in lung and bone marrow
Cause of Death	**Confirmed drowning (asphyxia due to water inhalation)**
Follow-up	Water sample collected from Gurda River for arsenic/chemical analysis; site flagged for earthen ramp installation

10. Two-Day Workshop Programme

Day / Time	Session	Resource Person
Day 1 Morning	Inauguration & Overview | CG Elephant Population & Mortality Statistics | Biological & Anatomical Aspects of Asian Elephants	Forest Officers, CG; Dr. Parag Nigam (WII)
Day 1 Afternoon	Non-Infectious Diseases in Asian Elephants | Infectious Diseases & Surveillance | Practical: PM Examination Techniques	Dr. Karikalan Mathesh (ICAR-IVRI); Dr. Tapendra Saini (WII)
Day 2 Morning	Sample Collection & Chain of Custody | Non-Infectious Pathology | Drowning & Electrocution Investigation	Dr. A. B. Shrivastav (NDVSU); Dr. C. P. Sharma (WII)
Day 2 Afternoon	Field Investigation Case Studies | Carcass Disposal & Site Remediation | Recommendations & Valediction	All resource persons; CG Forest Dept Officers

11. Key Recommendations & Priority Action Plan

Based on the 5-year mortality analysis, the following eight priority actions were identified by the expert panel as immediate interventions for the Chhattisgarh Forest Department:

#	Priority Action	Timeline	Responsible Agency
1	**Environmental Water Sampling:** Comprehensive toxicology sampling of Rabo and Panikshet dams and water bodies in Tamnar Range for arsenic and heavy metals	Immediate	CG Forest Dept + ICAR-IVRI
2	**Drowning Prevention Infrastructure:** Install earthen ramps and exit pathways at all identified drowning hotspots; priority: Ghargoda and Chhal ranges	1–3 months	CG Forest Dept + PWD

3	**Power-Line Audit:** Complete audit and rectification of illegal electric fencing and stray HV lines across all elephant corridors; special focus on Kharsia, Ghargoda and Dharamjaigarh	1–6 months	CG Forest Dept + CSPDCL
4	**Oct–Dec Enforcement Campaign:** Deploy anti-poaching/anti-fencing squads during harvest season with weekly intelligence gathering in conflict-prone villages	Seasonal	Forest Dept + Police
5	**Jan–May Calf Monitoring:** Intensive calf surveillance during dry season when calves are most vulnerable to drowning in drying water bodies	Seasonal	WII + CG Forest Dept
6	**Mandatory PM within 24 hrs:** Standard Operating Procedure mandating complete post-mortem examination with sample collection within 24 hours of elephant death, with digital reporting to State WL Office	Policy — immediate	PCCF (WL), CG
7	**Annual EEHV Surveillance:** Annual serological and PCR surveillance for Elephant Endotheliotropic Herpesvirus across all elephant herds in DH and RG Divisions	Annual	ICAR-IVRI + WII
8	**Quarterly Review Meetings:** Quarterly inter-departmental review of mortality data, action plan implementation, and hotspot management with representation from Forest Dept, Police, Revenue, and Power Department	Quarterly	PCCF (WL), CG — All Depts

Summary Statistics at a Glance

Metric	Value	Metric	Value
Study Period	2021–2026 (5 yrs)	**Total Elephant Deaths**	49
DH Division Deaths	27 (55%)	**RG Division Deaths**	22 (45%)
Electrocution Deaths	23 (47%)	**Drowning Deaths**	13 (27%)
Calf Deaths (0-2 yr)	23 of 46 (50%)	**Male Deaths**	23 of 45 (51%)
Peak Death Month	October (8 deaths)	**Peak Year**	2022-23 & 2025-26 (12 each)
Drowning: Calves	12/13 = 92%	**Arsenic Positives**	2 specimens (Tamnar, Dec 2025)
Human Deaths (HEC)	45 (DH: 36 | RG: 9)	**Pop. Growth (2001-26)**	24 → 451 (CAGR ~13%)

Conclusion: The 5-year mortality analysis clearly demonstrates that elephant deaths in Chhattisgarh are overwhelmingly anthropogenic and preventable. Electrocution (47%) and drowning (27%) together account for three-quarters of all deaths. Calf mortality is disproportionately high at 50%. The record mortality in 2025-26 is a critical signal requiring immediate multi-agency intervention. This workshop provides the technical foundation for standardised investigation, evidence-based reporting, and preventive action.